07 - Exercise Lab

Pattern Recognition and Machine Learning, Lab 7

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Today

Plan for today

Three problems

- maximum likelihood
- Parzen windows
- Neyman-Pearson
- Bayes with a reject option

All from **previous tests**.

Next week

- **Test 2** on 13 November
- at the start of the lab
- bring a **calculator**

test

Try each problem **yourself** first. Then click to reveal the results, Exercise 3 is solved step by step.

Exercise 1: maximum likelihood

Exercise 1: light bulbs

test_2019-eng.pdf

(8 points) Maximum Likelihood Estimation.

(a) (3 points) The reliability of light bulbs is defined by the probability density function $p(t) = \frac{1}{\theta} e^{-t/\theta}, t \in [0, \infty]$, where $p(t)\delta$ is the probability that a bulb fails in a short interval δ centered at time t. What is the maximum likelihood estimate of the reliability parameter θ if in an experiment with six bulbs the following lifetimes have been observed:

$x = \{t_0 = 0.0, t_1 = 1.0, t_2 = 2.3, t_3 = 2.7, t_4 = 6.0, t_5 = 12.0?\}$

(b) (3 points) If the test decribed above finished at time $t_{end} = 4.0$ two light bulbs would still be on, i.e. their time of failure would be unknown. What would an ML estimate for θ be in this case? Solve the ML estimation directly using probability of the light bulbs still functioning at time t_{end} for the uknown failure times (2 points). ~~Also do this using EM algorithm which will "substitute" for the unknown failure times the expectation of the failure time after t_{end}~~ (1 point).

(c) (2 points) The same problem, but the device for logging the failure times is not available. The only information we have is that at time t_{end}, G bulbs are still good and F bulbs have already failed. Find θ.

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c. $\hat{\theta}_{\mathrm{ML}} = \dfrac{T}{\ln \frac{F+G}{G}}$, for $T = 4, F = 4, G = 2$: $\hat{\theta}_{\mathrm{ML}} = \dfrac{4}{\ln 3} \approx 3.64$

Exercise 2: Parzen and Neyman-Pearson

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test_2021.pdf

(6 points) Parzen Windows and Neyman-Pearson.
There are two classes denoted 1 and 2. Let the training sets for the two classes be $X_1 = \{1, -1, 1, 2, 0\}$ and $X_2 = \{-1, 0, 1, 3\}$, respectively.

(a) (3 points) Use the training data X_1 to estimate the conditional density $p(x \mid 1)$ using the Parzen window method. Do the same for X_2 to estimate $p(x \mid 2)$. Employ kernel function $K(x, y) = k(x - y)$, with $k(z)$ defined as:

$$\begin{aligned} k(z) &= 1/h \quad \text{for } |z| \leq h/2 \,, \\ k(z) &= 0 \quad \text{for } |z| > h/2 \,, \end{aligned}$$

with $h = 2$. Plot the estimated densities. Find values of $p(x \mid 1)$ and $p(x \mid 2)$ for $x = 0.5$.

(b) (3 points) Consider class 2 as dangerous. Construct the Neyman Pearson strategy for the following setting: overlooked danger $< 40\%$. For this strategy, what are the errors ϵ_1 and ϵ_2 for class 1 and 2, respectively?

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Even better: add a piece with $r = 0.8$ and mass $\hat{p}(x \mid 2) = 0.025$ to fill $\epsilon(2)$ up to 0.4. There are **infinitely many** such pieces, e.g. $[2, 2.2)$ or $[-0.1, 0)$, all give $\epsilon(1) = 0.4 - 0.8 \cdot 0.025 = 0.38$:

$$q(x) = \begin{cases} 1 & 0 \le x < 2.2 \\ 2 & \text{otherwise} \end{cases} \qquad \epsilon(1) = 0.38, \quad \epsilon(2) = 0.4$$

Exercise 3: Bayes with a reject option

Exercise 3: two normal classes

test_2019b-eng.pdf

(6b) Bayesovská klasifikace. Uvažujte dvě třídy 1 a 2, které mají obě normální rozdělení, $N(\mu_1, \mathbf{C}_1)$ a $N(\mu_2, \mathbf{C}_2)$, a stejné apriorní pravděpodobnosti. Parametry rozdělení jsou:

$$\mu_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \mathbf{C}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \mu_2 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \mathbf{C}_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

(a) (2b) Najděte klasifikátor, který minimalizuje chybu klasifikace.

(b) (2b) Reject option. Uvažujte, že je možné rozhodnout nejen '1' nebo '2', ale i 'nevím'. Za rozhodnutí 'nevím' se zaplatí 0.1 jednotky, za chybu 1 jednotka (a správné rozhodnutí je zdarma). Pro které hodnoty $\mathbf{x} \in \mathbb{R}^2$ Bayesovská optimální strategie rozhodne 'nevím'?

(c) (2b) Najděte klasifikátor $q\colon \mathbb{R}^2 \to \{1, 2\}$, který minimalizuje Bayesovské riziko, pričemž ztrátová funkce je $W(1, 2) = 2$ a $W(2, 1) = 1$, tj. za chybnou klasifikaci objektu třídy 1 jako 2 se zaplatí dvakrát více než za opačnou chybu.

Derivation: (a) minimal error

Equal priors, decide by the log ratio of the likelihoods:

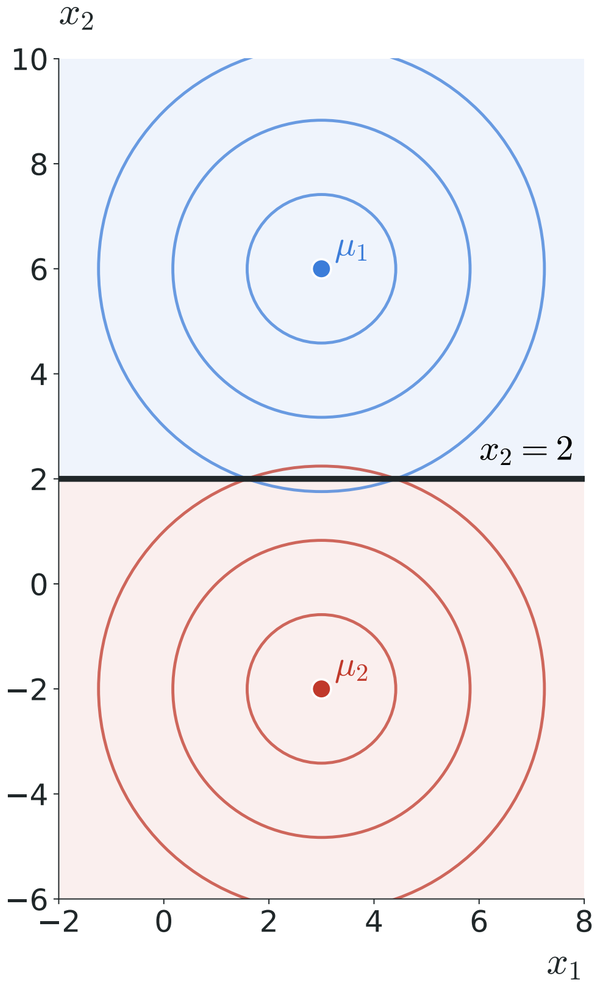
$$a(x) = \ln \frac{p(x \mid 1)}{p(x \mid 2)} = \frac{\|x - \mu_2\|^2 - \|x - \mu_1\|^2}{4}$$

The terms with x_1 cancel:

$$a(x) = \frac{(x_2 + 2)^2 - (x_2 - 6)^2}{4} = 4x_2 - 8$$

$$q(x) = \begin{cases} 1 & x_2 > 2 \\ 2 & x_2 \leq 2 \end{cases}$$

A line halfway between the means.



Derivation: (b) reject option

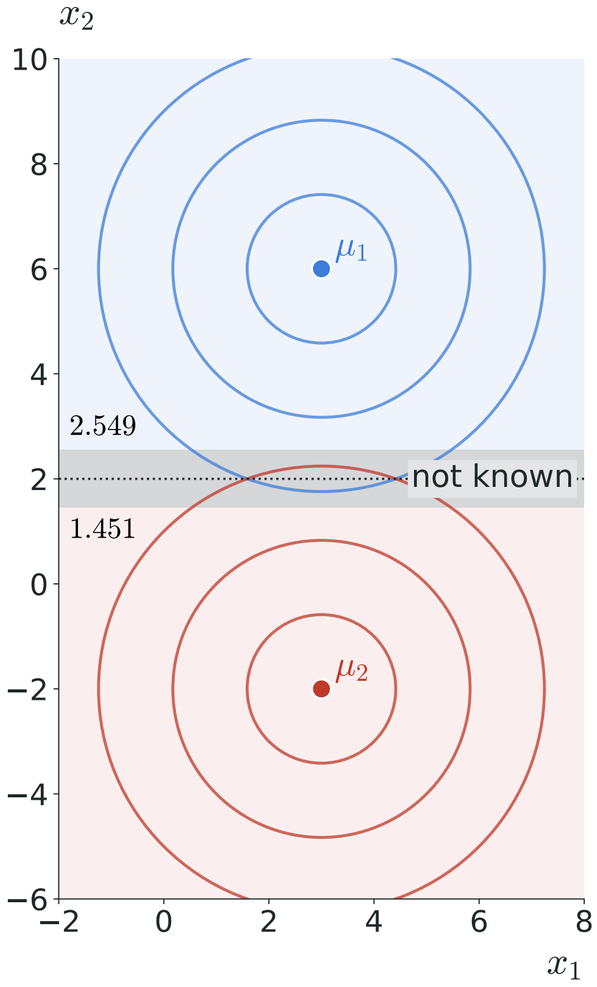
Bayes with "not known" (Lab 3): decide the most probable class if

$$1 - \max_k p(k \mid x) < 0.1 \quad \Leftrightarrow \quad \max_k p(k \mid x) > 0.9$$

From (a): $p(1 \mid x) = \sigma(a(x))$ and
$\sigma(a) > 0.9 \Leftrightarrow a > \ln 9$

$$\text{not known} \;\Leftrightarrow\; |4x_2 - 8| < \ln 9 \;\Leftrightarrow\; |x_2 - 2| < 0.549$$

$$q(x) = \begin{cases} 1 & x_2 \geq 2.549 \\ \text{not known} & 1.451 < x_2 < 2.549 \\ 2 & x_2 \leq 1.451 \end{cases}$$



Derivation: (c) asymmetric loss

Risk of each decision:

$$R(1 \mid x) = W(2,1)\, p(2 \mid x) = p(2 \mid x)$$
$$R(2 \mid x) = W(1,2)\, p(1 \mid x) = 2\, p(1 \mid x)$$

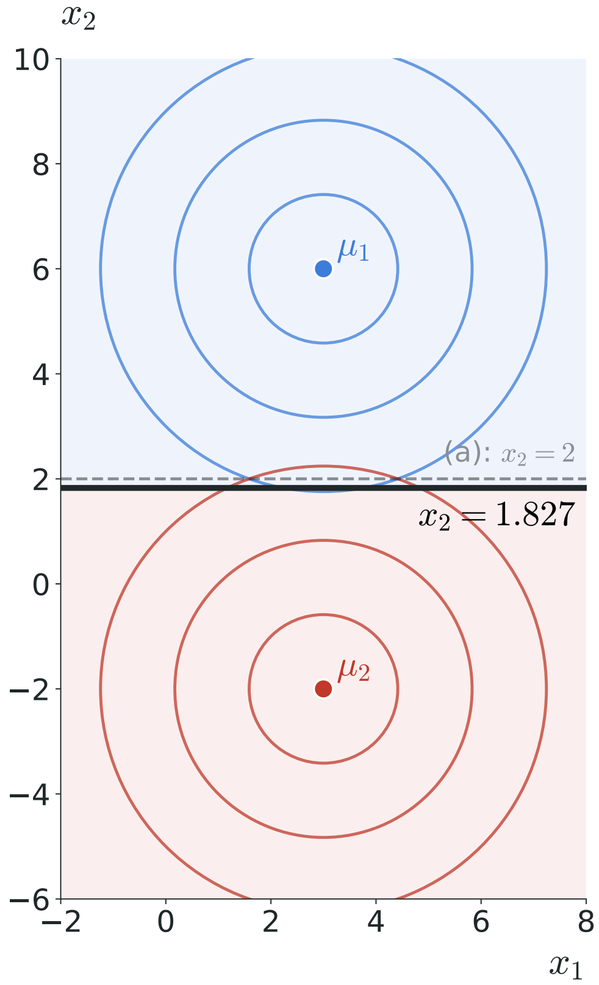
Decide 1 if

$$p(2 \mid x) < 2\, p(1 \mid x) \;\Leftrightarrow\; a(x) > -\ln 2$$

so $x_2 > 2 - \frac{\ln 2}{4} \approx 1.827$

$$q(x) = \begin{cases} 1 & x_2 > 1.827 \\ 2 & x_2 \le 1.827 \end{cases}$$

An error on class 1 is expensive: the boundary moves **towards** μ_2.



Take-away for the lab

- ☑ **Censored data**: a bulb that still works contributes $P(t > T)$, not a density
- ☑ **Only counts**: a binomial likelihood, as for the socks
- ☑ **Parzen** with a box kernel: a step function, count the boxes
- ☑ **Neyman-Pearson**: sort by $r(x)$, add regions until the limit
- ☑ **Equal covariances**: linear log odds, reject and loss shift the threshold


"That's all Folks!"